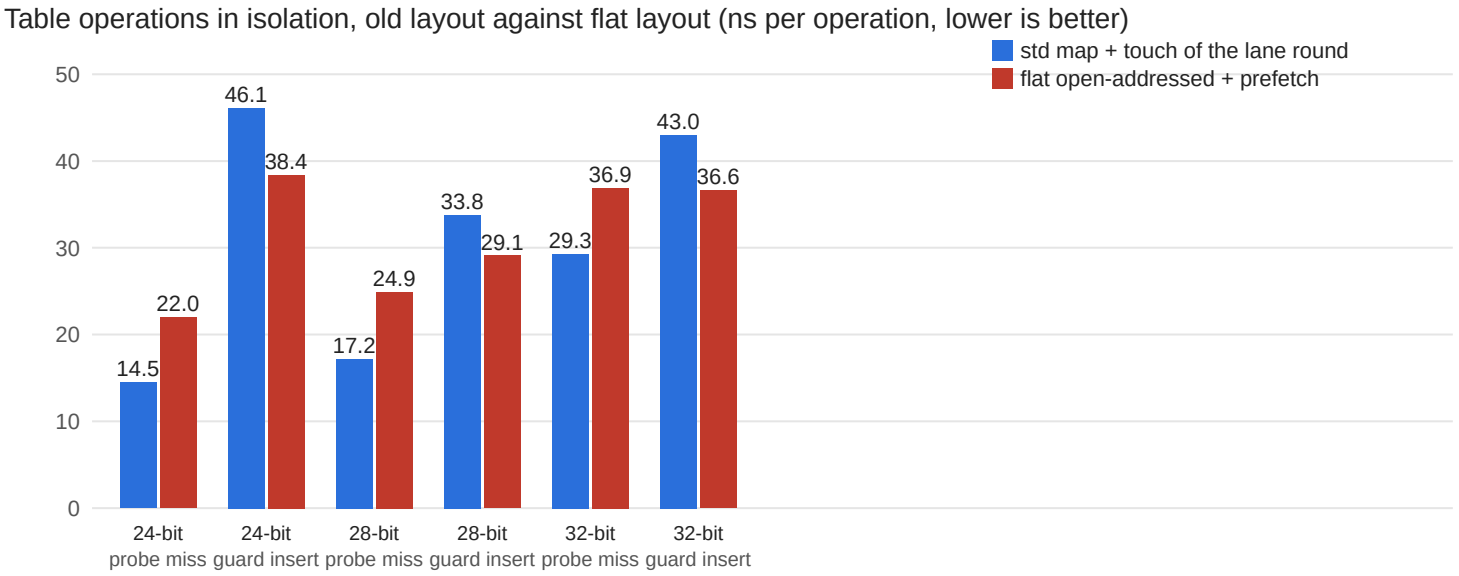
The hash-table floor of the prime IC relation trial: a null result (2026-10-29)

Lead `ICP-prime-ic-constant-engineering`. The wide-batching study (`research/ic_wide_ext_20261028`) measured the target guard insert and the oracle probe at about 137 ns of a ~318 ns trial and called it the floor. A one-region open-addressed pair table (one 64-bit word per entry) and an open-addressed target guard, with real prefetches for a lane round, were built to test whether that floor can be lowered with identical output. Results below are the two operations timed in isolation on the real tables of the tuned bases (median of seven passes, one million uniformly random keys; almost every probe misses, as a target's abscissa does). The paired ecbench measurement was not run: another job held the benchmark lock for the whole session.



bits	base	entries	probe, miss (ns)				probe, hit (ns)		guard insert (ns)	
			map	flat	map + touch 16	flat + prefetch 16	map	flat	map + touch 16	flat + prefetch 16
24	256	65250	10.7	24.7	14.5	22.0	4.8	14.6	46.1	38.4
28	512	261846	13.4	39.0	17.2	24.9	5.9	18.7	33.8	29.1
32	1024	1048201	22.4	60.7	29.3	36.9	8.0	34.6	43.0	36.6

Findings

- **The flat pair table is slower, not faster.** A miss costs 2.3 to 2.9 times as much in the one-region word table as in the std map (24.7/39.0/60.7 ns against 10.7/13.4/22.4 ns), and still 1.3 to 1.5 times as much once the lane round touches the map and prefetches the word table (22.0/24.9/36.9 ns against 14.5/17.2/29.3 ns). The std map's miss path reads one control-byte group and compares a 7-bit tag; the word table scans linear-probe slots (load up to 3/4) in a table that is four times larger than the control bytes.
- **The open-addressed guard saves 5 to 8 ns of insert time** (46.1 to 38.4, 33.8 to 29.1, 43.0 to 36.6 ns), about 2% of a 318 ns trial. That is not enough to land without an end-to-end measurement, and the end-to-end measurement was blocked.
- **The floor is smaller than the diagnosis said.** The two operations together are 51 to 72 ns in isolation (map + touch probe plus guard), against 137 to 140 ns in the diagnosis. The diagnosis times each part of the loop body with a pair of clock reads; the difference is consistent with that cost plus cold-cache effects between operations, not with the tables. I did not separate the two.
- **Verdict.** The table-layout item of the lead closes as measured: there is no layout of this shape that lowers the relation phase, and the part of the floor that is table work is under a quarter of a trial.

Not done

- The paired old/new ecbench sessions and the replay certificate. A job queue of multi-hour sessions held `/tmp/crypto-bench.lock` from before this study started; the sessions I launched were refused ("another benchmark holds the lock") and I did not bypass the lock.
- Nothing in this branch changes a production code path: the flat table and the guard set exist only in the test module, with equivalence tests against the map forms.